

Compressive strength of concrete material using machine learning techniques

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ABSTRACT

Significant efforts have been made to improve the strength of concrete by utilizing industrial waste like Fly Ash as a partial replacement of cement in the concrete. However, predicting the compressive strength of concrete is one of the challenging tasks since it is affected by several factors such as the shape and size of aggregates, water-cement ratio. The paper presents a study on the various investigation of machine learning (ML) algorithms to estimate the compressive strength (CS) of concrete containing fly ash (FA). The research also aims to compare the accuracy of different ML models, including non-ensemble models (Multiple Linear Regressor, Support Vector Regressor) and ensemble models (AdaBoost Regressor, Random Forest Regression, XGBoost Regressor, and Bagging Regressor), in predicting CS with a focus on identifying the most accurate estimation method. For this purpose, a dataset of 633 experimental results with a wide range of CS values, ranging from 6.27 MPa to 79.99 MPa, was collected from existing literature and validated using statistical analysis. The primary input parameters for the ML models included the quantities of cement, fine aggregate (FA), coarse aggregates (CA), fly ash (FA), water content, percentage of superplasticizer, and curing days, with the CS of concrete as the output. Performance evaluation was conducted using several performance indices, such as MAE, MSE, R^2 , MAPE, RMSE, and a20-index, to assess accuracy and reliability. The comparison reveals that XGBoost Regressor is the most reliable model, demonstrating the highest coefficient of determination (R^2) of 0.95, a-20 index of 0.913, and the lowest RMSE value of 3.06 MPa and MAE of 2.13 MPa, while the Multiple LR model is the least accurate method with R^2 equal to 0.52, a-20 index of 0.433, RMSE of 9.40 MPa and MAE of 7.68 MPa. Additionally, to provide deeper insights into the relationship between input parameters and CS, sensitivity and parametric analysis were employed, enabling a comprehensive understanding of the impact of fly ash and other parameters on CS prediction. From the study, it was observed that the age of the concrete is the most essential feature, followed by cement and water, with information gain values of 32.91, 23.50, and 15.10, respectively. The study highlights the effectiveness of machine learning techniques, particularly the XGBoost Regressor, in accurately estimating the compressive strength of concrete. Furthermore, it offers researchers a faster and more cost-effective means of evaluating the effect of fly ash and other factors on CS estimation, avoiding the need for time-consuming and costly experimental studies.

1. Introduction

Concrete is the second most popular construction material after water due to its global availability of raw materials, ease in processing and handling, and its inherent ability to switch from a fluid state, making it able to fill in a mold to a solid state, capacity to resist imposed

structural load (Wangler et al., 2019). Cement is one of the main constituents of concrete among sand, coarse aggregate, water, and other materials. The manufacturing process of Portland cement involves the calcination of calcium oxide, which results in the production of CO_2 (Kajaste and Hurme, 2016), (Barcelo et al., 2014) and has a serious impact on the environment. To reduce the production of CO_2 , concrete

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should be produced using as much recycled or waste material as possible. (Batayneh et al., 2007). Moreover, the reduction in carbon footprint can be attained by substituting the cement with alternative cementitious materials such as rice husk ash, fly ash (FA), blast furnace slag, Metakaolin, and Silica Fume to decrease the impact of concrete on the environment (Gartner, 2004). (Phul et al., 2019), (Shubbar et al., 2018), (Czarnecki et al., 2021). Several studies have shown that the adoption of FA as a partial substitution for Ordinary Portland Cement (OPC) has been advantageous in reducing carbon footprints and will enhance the ultimate strength, workability, and durability of hardened concrete compared to conventional concrete (Nath and Sarker, 2011), (Bahedh and Jaafar, 2018). Kavitha et al. (2020) studied the effect of FA on the mechanical as well as sulphate resistance test and acid attack test of high-performance concrete. They observed that the optimum percentage of fly ash is 15% to enhance the durability and compressive (CS), and tensile strength of concrete and for the social and economic benefit (Kavitha et al., 2020). Likewise, Szczesniak et al. (2020) investigated the effect of FA on the mechanical and physical properties of concrete comprising cement with slag and found improvement in the tightness and chloride ion penetration resistance and high strength over the maturity of concrete. Moreover, Alumina and silica are the main ingredients in fly ash. The impact of SiO_2 and CaO in FA on the CS from soft computing models for different concrete mix proportions was examined by Jaf et al. (2023). Ali et al., 2022 (Ali et al., 2022) investigated the effect of Nanosilica content (NS%) on the CS of concrete with various mix proportions and sizes of specimens using different ML models. The research work showed that the full quadratic (FQ) model exhibited the most accurate result with better R^2 values and lower error in prediction. It is commonly known that different materials have different compositions depending on their compressive strength range, such as those with compressive strengths between 10 and 70 MPa and higher than 70 MPa. The research of (Ali et al., 2022) compared the model performance for various strength ranges like 20–55 MPa, 55–70 MPa and 70–100 MPa with the help of root mean square error (RMSE), coefficient of determination (R^2), Scatter Index (SI), and mean absolute error (MAE). The FQ model performed the best, especially when it came to predicting the CS of the concrete with nanosilica and had a strength between 70 and 100 with R^2 equal to 0.89. However, the application of the model is more accurate in the 20–55 MPa range due to the bigger size of the data points; the R^2 score for this range was 0.82.

The compressive strength (CS) of concrete is the basic component adopted for the strength and durability design and estimation of the service life of the structures in civil engineering projects. The CS of concrete is the vital parameter as a majority of linear and nonlinear modeling parameters of the concrete can be obtained from the CS of the concrete (Paudel et al., 2021), (Tanapornraweekit et al., 2022), (Paudel, 2023). The CS of concrete can be adopted to calculate several mechanical properties for numerical simulations (Paudel et al., 2021a), (Paudel et al., 2021b), (Paudel and Bhusal, 2021). However, the CS of the mixture is greatly influenced by factors including water-cement ratio, shape and size of the aggregate, and composition of admixtures (Cotsovos and Pavlović, 2008). Therefore the prediction of compressive strength (CS) using laboratory tests is challenging, uneconomical, and turns out to be time-consuming. The regression analysis approach (Chopra et al., 2014), (Chithra et al., 2016) is thus another method for determining the CS of concrete to overcome the drawbacks of the traditional experimental procedure. Unfortunately, concrete is a complex nonlinear material, and the accuracy of the empirical regression method is influenced by the number of independent variables. With the advancement of artificial intelligence, attempts have been made to estimate the CS of concrete using machine learning (ML) techniques while taking into account the complexity and randomness of concrete materials (Khan et al., 2021), (Ahmad et al., 2021), (Naderpour et al., 2018), (Topcu and Saridemir, 2008), (Asteris et al., 2021a), (Asteris and Mokos, 2020) (Kostić and Vasović, 2015). The machine learning model is capable of obtaining a solution to highly nonlinear complex problems

with better efficiency by linking the provided input to the required output via a mapping function which is developed using pattern recognition techniques and multivariable regression.

Several researchers have investigated the machine learning application for the estimation of the CS of concrete. A study on predicting the CS of recycled aggregates was performed by Liangtao et al. based on an artificial intelligence algorithm (Bu et al., 2021). The authors adopted a dataset to train Artificial Neural Network (ANN) model from the experiments. The neural network comprised eight input variables with two hidden layers and was trained with 88 datasets obtained from the experiment. The sensitivity analysis was performed on the input variables to obtain their influence in the prediction of CS of recycled aggregate concrete (RAC). The study highlighted that the adopted model could predict the CS of RAC with better accuracy. Yu et al. (2018) used a swarm-based algorithm called cat swarm optimization (CSO) (Chu and Tsai, 2007) to predict the CS of high-performance concrete (HPC). The estimate of CS of HPC requires an excessive trial mix which results in excessive waste of cost, time, and materials, so a reliable prediction of the strength based on the machine learning approach can be a better option. Yang et al. (Yu et al., 2018) developed a model with coarse and fine aggregates, contents of cement, water, fly ash (FA), curing age, superplasticizer, and blast furnace slag as input parameters, and the output of the trained model is CS of HPC. Several indices like relative root mean square error (RRMSE), mean absolute error (MAE), error to signal ratio (ESR), standard deviation (σ), the slope of the regression line (SRL), coefficient of determination (R^2), index of agreement (IA) and mean absolute percentage error (MAPE) were used to check the performance of the adopted ML algorithms. The authors concluded that the adopted algorithms showed good accuracy in terms of the above listed indices and hence can be used to predict the CS of the HPC.

A.J. Tenza-Abril et al. (2018) used ultrasonic pulse velocity and ANN to estimate the CS of segregated lightweight concrete (SLC). A comprehensive experimental dataset with 640 data points was adopted to select the suitable ANN model. The model was adopted to investigate the contribution of compressive strength, concrete density, and longitudinal wave results on segregation phenomena and the influence of segregation on Lightweight aggregate concrete (LWAC) compressive strength. The authors concluded that the adopted ANN model provides a rapid and precise estimate of the CS of segregated LWAC. Feng et al. (2020) used an adaptive boosting ML algorithm to forecast the CS of concrete. A comprehensive dataset of 1030 datapoints in terms of CS of concrete and with input parameters comprising of aggregates (coarse/fine), cement, water, curing time, etc., was adopted to construct the model. A unique approach, namely the adaptive boosting (AdaBoost) algorithm, was adopted to find the relationship between the input and output data points (Wu et al., 2008). Moreover, several weak learners (which have small predicting errors) were integrated to form a strong learner. Also, the influence of the choice of a weak learner, the training data sets, the number of input parameters, and the effect of the sensitivity in the adaptive boosting algorithm were assessed. The study concluded that the AdaBoost model could predict the CS of different mix proportions accurately and can be adopted to check the accuracy of the predicted CS can satisfy the target strength requirements. Song et al. (2021) used an ML algorithm to predict the CS of concrete with FA admixture. In the ANN network, cement binder, FA, coarse/fine aggregate, water, superplasticizer, and days were provided as input for the prediction of CS as an output parameter. The validation of test data was performed by root mean error (RME), k-fold cross-validation using R^2 , and Root mean square error (RMSE), and finally, the model performance was evaluated based on statistical checks. The study concluded that the ML approach provided a strong correlation between the predicted and targeted results. In summary, several studies have been carried out so far using ML techniques, including the prediction of the shear strength of concrete beams (Armaghani et al., 2019), (Asteris et al., 2019), prediction of the CS (Asteris et al., 2021a), (Armaghani and Asteris, 2021), and CBR ratio of soil (Bardhan et al., 2021), and axial load carrying

capacity of concrete-filled steel tube columns (Bardhan et al., 2022), to name a few. Since last decade several research works have been carried out to forecast the compressive strength of environmentally friendly concrete using waste glass powder, granulated blast furnace slag (GGBS), eco-friendly self-compacting concrete from recycled plastic aggregates and ash from industrial waste innovative computing methods, such as artificial neural networks and nonlinear regression models, are used (Ahmad et al., 2023), (Ahmed et al., 2022), (Faraj et al., 2022). Additional research works include the prediction of the flowability of concrete (Mahjoubi et al., 2022), strength and analysis of self-compacting concrete (Rajakarunakaran et al., 2022). A summary of the different ML algorithms adopted in past literature as well as in the current study is presented in Table 1.

2. Research objectives

The study presented herein extends the efforts made by previous researchers through the use of ML algorithms to estimate the CS of concrete containing FA. Additionally, it should be noted from Table 1, that limited research has been conducted using both non-ensemble and ensemble models. Hence the current study was conducted using both types of models to account for a wide range of compressive strength (CS) of concrete data, ranging from 6.27 MPa to 79.99 MPa. The research also aims to evaluate the accuracy (reliability) of different ML algorithms, namely Multiple Linear Regressor, Support Vector Regressor, AdaBoost Regressor, Random Forest Regression, XGBoost Regressor, and Bagging Regressor and the most optimal method to predict the CS. Performance evaluation of the models was conducted using several performance indices, such as MAE, MSE, R^2 , MAPE, RMSE, and a20-index, to assess accuracy and reliability. Additionally, this study uses sensitivity and parametric analysis to establish the relationship among the input parameters to estimate the CS of concrete. This will enable the researcher to evaluate the effect of fly ash and other parameters on predicting CS quickly and more economically than costly experimental studies.

Table 1
Machine learning (ML) models adopted to estimate the CS.

ML Algorithm name	Dataset	Year	Ref.
Gradient boosting regression tree (GBRT)	1030	2023	Li et al. (2023)
Ensemble and non-ensemble supervised ML approach	471	2022	Song et al. (2022)
Genetic expression programming (GEP) and ANN	310	2022	Al-Hashem et al. (2022)
Firefly Algorithm (FA) and Random Forest (RF)	225	2022	Huang et al. (2022)
Extreme Gradient Boosting (XGBoost), RF, and Support Vector Machine (SVR)	1030	2022	Wang et al. (2022)
GEP	357	2020	Aslam et al. (2020)
Ensemble and Individual (GEP, DT, and Bagging)	270	2021	Ahmad et al. (2021)
Data envelopment Analysis	114	2021	Balf et al. (2021)
Multivariate	21	2020	Busić et al. (2020)
Adaptive neuro-fuzzy inference system	7	2020	Al-Mughanam et al. (2020)
Intelligent rule-based enhanced multiclass support vector machine and fuzzy rules	114	2019	Selvaraj and Sivaraman (2019)
RF	131	2019	Zhang et al. (2019)
Multivariate adaptive regression spline	114	2018	Kaveh et al. (2018)
ANN	114	2017	Belalia Douma et al. (2017)
Non-ensemble: Multiple LR and SVR Ensemble: AdaBoost regressor, Random forest regression, XGBoost regressor, and Bagging regressor	Current study		

3. Background on ML algorithms

Machine learning (ML) mathematical models are inspired by biological principles of human brain structure. The models are capable of being trained, computing, and adapting themselves from the provided range of datasets and finally can predict the new dataset. The machine learning model consists of input and output layers, with input and output variables, respectively. The two layers are connected via one or several hidden layers, which perform the required calculations and identify the hidden pattern and relationship between the provided input datasets. The abovementioned neurons interconnect the layers via activation functions (usually nonlinear) such as ReLu, Step, Sigmoid, Tanh, etc. The activation of neurons in any layer is calculated with the help of these activation functions. Finally, the nonlinear output for the neurons is provided and is fed to other neurons located in several other layers by these functions. A typical neural multilayer perceptron employing input, sum function, activation function (sigmoid), and output is shown in Fig. 1.

To predict the new dataset, the model utilizes several neurons (processing units) which are linked together to form a network and process information. Moreover, the neurons are linked with each other through weights and biases (deviation between weights), and these affect the accuracy of the machine learning model (Aljarah et al., 2018), (Whitley et al., 1990). The number of hidden layers distinguishes the network into a simple ANN (single hidden layer) and a deep neural network (several hidden layers) (Deng et al., 2013). The adoption of multiple hidden layers enhances the capability of the ANN to identify the interrelationship between the provided input and required output variables and improves the accuracy of the developed model (Schmidhuber, 2015). A neural network's basic structure consists of a single input, two hidden layers, and a single output layer, as shown in Fig. 2.

In the current study, individual ML models and ensemble models with weak learners were adopted to model the strength of concrete which is discussed in subsequent sections.

3.1. Non-ensemble models

Two non-ensemble machine learning models, namely Multiple LR and support vector regression (SVR), were adopted in the current study.

a. Multiple Linear regressions (MLR)

The regression models estimate the level of correlation and determine the dependence among the input and output variables. The linear regression (LR) models are mostly fitted by the least square approaches. The MLR evaluates the correlation between the response (dependent) from several independent variables and produces a straight-line relationship (Neter et al., 1996). The expression of MLR is shown in Eq. (1).

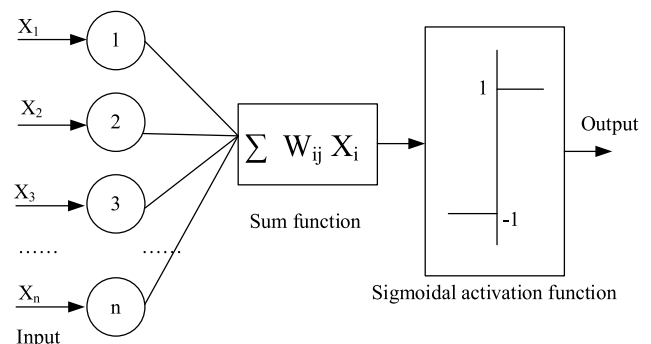


Fig. 1. A typical neural model employing the sigmoid function.

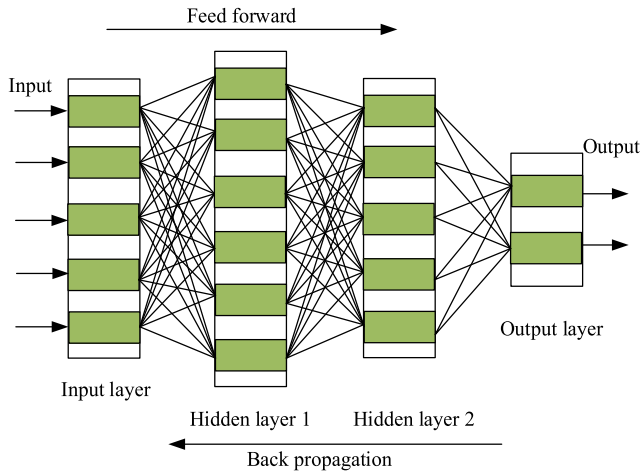


Fig. 2. Machine learning execution process.

$$Y = a_0 + \sum_{j=1}^m a_j X_j \quad (1)$$

where Y is the output of a model, X_j 's are the input variables to the algorithms, and $a_0, a_1, a_2, \dots, a_m$ are partial regression coefficients.

b. SVR

The support vector machine model is a typical data mining model, and when it is used for regression problems, it is called support vector regression (SVR). This method was developed in 1995 by Cortes and Vapnik (Cortes and Vapnik, 1995) and is currently used for classification, prediction, and regression. The SVR is a supervised learner and is used mostly for input-output analysis as it can handle problems of nonlinear regression, small databases, and high input dimensional spaces effectively. This approach converts input space into high-dimensional space using a nonlinear transformation. Then after matching the substantially high-dimensional space, the nonlinear activation operations are employed. Here, there is a lot better match because the input parameters are substantially distinct from the initial data set. The linear function in space $f(x, w)$, can be estimated from the expression as shown in Eq. (2).

$$f(x, w) = \sum_{i=1}^n w_i g_i(x) + b \quad (2)$$

3.2. Ensemble models

Ensemble models are the learning algorithm that constructs a set of classifiers whose individual decisions are combined by taking the weighted vote of their predictions (Dietterich, 2000). The Bayesian averaging method was the original ensemble method, and later on, several algorithms like boosting, Bagging, and error-correcting output coding were used widely in machine learning. These models can obtain highly accurate classifiers by combining less accurate ones. The ensemble model works better than a single classifier due to three fundamental reasons, namely statistical, computational, and representational (Dietterich, 2000). Four ensemble machine learning models, namely AdaBoost, Random Forest, XGBoost, and Bagging were adopted in the current study. The Bagging and boosting ensemble learning algorithms are designed for supervised learning. Fig. 3 describes a binary dataset consisting of white and black circles, intending to build a model that can correctly classify these data points. Model 1, a weak learner, is fitted on the training dataset, but it makes mistakes in classifying some points. Boosting is used to improve the performance of Model 1 by adding Model 2 and, subsequently Model 3, which focus on correcting the shortcomings of the previous models. However, adding too many models might lead to overfitting, and the number of models to be added is a hyperparameter in boosting.

a. AdaBoost

Boosting is one of the popular and most widely used machine learning algorithms and was suggested by Freund and Schapire in 1990 (Schapire, 1990). Freund and Schapire 1995 introduced adaptive boosting (AdaBoost) to combine the weak classifier gained during training into a strong one and to adjust the training set to conduct the training on the weak classifier (Chengsheng et al., 2017).

b. Random forest (RF) regression

An RF model is mostly used due to its better performance in regression and classification problems. It is supervised learning and uses an ensemble learning approach for regression. It is an ensemble of decision tree models under the bagging framework and consists of root nodes, child (or split) nodes, and leaf nodes (i.e., these have no child nodes). Each node in the algorithm has a criterion based on the provided input features. The algorithm determines whether something is true or untrue based on the criteria as it moves from the root to the leaf node. The RF algorithm was developed by L. Breiman, 2001 (Breiman, 2001) by combining the random selection of features (Ho, 1995) and bootstrap aggregation (Breiman, 1996).

c. XGBoost

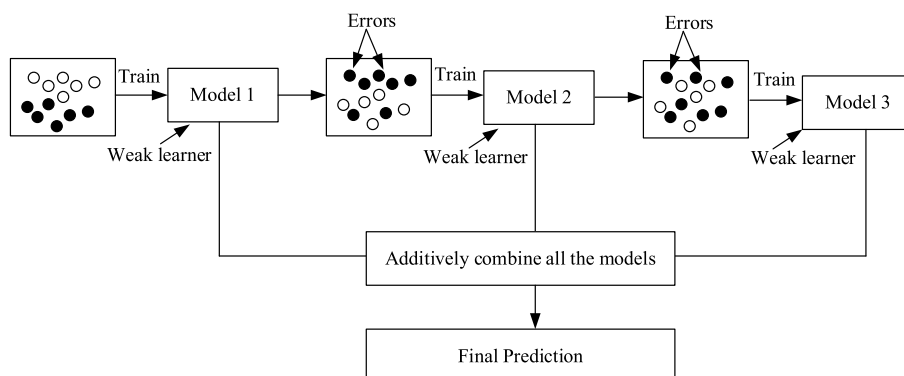


Fig. 3. Generic boosting framework.

XGBoost is a different, extensively used scalable ML algorithm for tree boosting. The scalability of XGBoost is due to several important features like parallel and distributed computing, a novel tree learning algorithm for handling sparse data, and algorithmic optimizations (Chen and Guestrin, 2016). In the current study, XGBoost was selected due to the several relevant features such as regularization, second order gradient for faster convergence, sparsity aware split finding, use of stochastic GD for introducing diversity and reducing overfitting, shrinkage for reducing overfitting and a series of system features: parallelization, cache optimization for efficiency and scalability.

d. Bagging

Bagging is a method that uses bootstrap sampling to create multiple datasets from a single dataset by sampling with replacement. This creates independent samples, which are used to train multiple models, as shown in Fig. 4. The final model is obtained by averaging these models for regression or using mode for classification. Bootstrap samples have some duplicates, with only 63% of the data being unique. Bootstrap aggregating, commonly referred to as Bagging, is the most widely adopted ensemble method developed by Breiman (1996). This technique randomly selects a sample of the original data and aggregates each regressor's prediction depending on the voting or averaging procedures to get at final forecasting.

This kind of meta-estimator can greatly reduce the variation resulting from the black-box estimation technique by developing a randomization procedure in their prediction formulation. Furthermore, BGR performs better when dealing with sophisticated algorithms like fully completed decision trees by figuring out a way to reduce overfitting. Fig. 5 illustrates the schematic diagram of the methodology adopted in the current study. The specific steps are as follows.

- 1) Data preprocessing: For the research work, 633 datasets were collected from the existing research database, and the data were analyzed to explore the statistical properties based on histograms and correlation matrix.
- 2) The non-ensemble and ensemble ML models were adopted, as shown in Fig. 5, and the results obtained from the analysis were evaluated from different performance metrics.
- 3) The dataset was further divided into training and testing data, and the k-fold cross-validation was carried out. Additionally, the accuracy of the model was checked based on performance indices, and the sensitivity analysis was carried out.

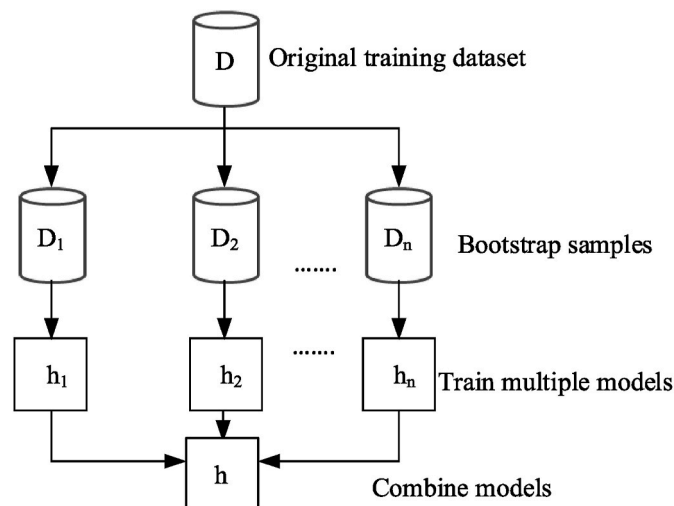


Fig. 4. Generic bagging framework.

- 4) Finally, parametric and sensitivity analysis was carried out to further analyze the data type and explore the importance/contribution of each input feature for the corresponding output

The detail of the adopted database and the results are explained in Section 3 and Section 4, respectively.

4. Description of database

To construct a prediction model and perform statistical analysis, either experimental laboratory test results or a collection of a database from already published literature can be applied. In the present study, a total of 633 data sets on the CS of fly ash-based concrete were gathered from a literature survey of past journal papers (Song et al., 2021), (Song et al., 2022), (Mahajan and Bhagat, 2022). The analysis of data was carried out based on seven characteristics. These seven characteristics were utilized as input parameters in the order of cement, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, curing days, and compressive strength as output parameters, and their relationship is illustrated in Fig. 6, which describes the relative frequency distribution of individual input and output parameters.

4.1. Statistical description and correlation analysis of data

The statistical description of the adopted dataset can be observed in Table 2. It contains the range of minimum and maximum values, standard deviation (S.D), and average values of the adopted input and output variables. To develop the best predictive model, the proposed expression in the current study should be used within the specified range of parameters. The statistical analysis illustrates that the adopted dataset covers a large range of ingredients, and S.D. depicts the distribution of the dataset along the mean values.

The distribution improves with the increase in SD values. The histogram of different input variables is shown in Fig. 7. According to the histogram, as shown in Fig. 7, it is obvious that the range compressive strength of all the collected datasets is 6.27–79.99 MPa. It means that the proposed model can be applied to estimate the CS of the samples lying in the aforementioned range. The figure clearly shows that the variables used in the database cover a wide range which proves the reliability of the collected database. Hence, compressive strength can be accurately predicted by the proposed model based on such a dataset.

The input variable which is not related to the CS of the concrete should be excluded from the dataset, which is obtained by correlation analysis. Hence, the process of determining how two or more variables are correlated to each other is known as correlation analysis. Among the commonly used correlation coefficients, Pearson correlation coefficient, Spearman correlation coefficient, and Kendall correlation coefficient, the first method is the most popular among researchers (Williams et al., 2020). The Pearson correlation coefficient ρ_{xy} is a ratio of covariance of two variables $\text{cov}(X, Y)$ to the product of their standard deviations (σ_X, σ_Y), given by Eq. (3).

$$\rho_{xy} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}} \quad (3)$$

Where $\bar{x}$ and $\bar{y}$ are the mean of two parameters X and Y.

The range of Pearson correlation coefficient ρ_{xy} lies within $[-1, 1]$. The high values of ρ_{xy} denote that the parameters are related and have a high effect on the output parameter while $\rho_x = -1$ means that variables are not directly correlated i.e., the relationship between the variables is inverse. Also, if $\rho_x = 0$, it should be noted that it does not mean variables are not correlated, they can have a nonlinear correlation. It represents parameters X and Y are linearly independent since the Pearson correlation coefficient is capable of showing only a linear relationship.

According to Fig. 8, cement and curing days have a high impact on

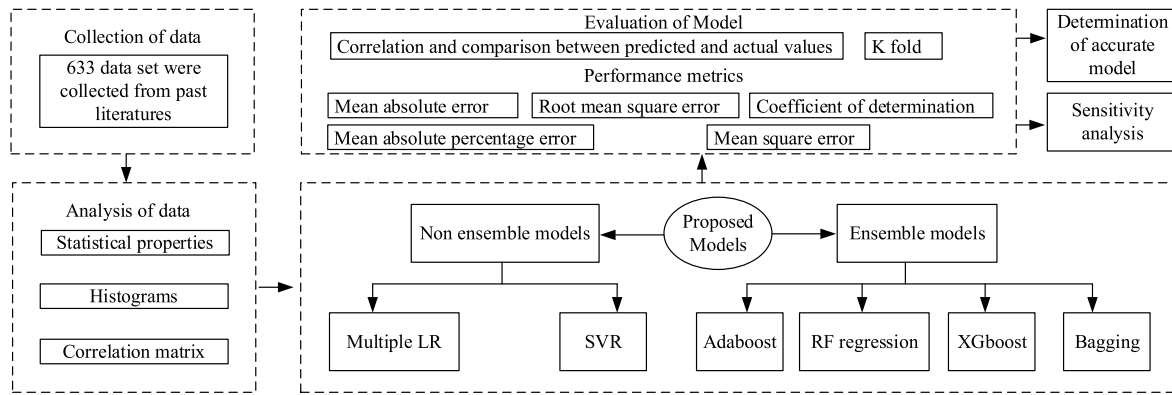


Fig. 5. Flowchart of the methodology of machine learning adopted in the current study.

the compressive strength of concrete, shown by their high values of the Pearson correlation coefficient, while the compressive strength of concrete is least affected by coarse aggregate. Fig. 8 also illustrates cement, superplasticizer, and age of the concrete are positively correlated with compressive strength, and the remaining other parameters are inversely correlated with the strength of the concrete, which denotes compressive strength of concrete decreases with the increase in water, fly ash, fine aggregate, and coarse aggregate. Also, there is no uncorrelated feature. Hence, all seven input parameters can be utilized to predict the CS of the concrete.

4.2. Data normalization

If the input scale varies, some machine learning models do not perform effectively. Table 2 shows that the range of cement is between 135 and 540 kg/m³, whereas the range of superplasticizers is between 0 and 28.2 kg/m³. This indicates that the magnitudes of the various input features vary. Data normalization, also known as rescaling, is the process of bringing all of the input variables' magnitudes to the same scale. It is accomplished using the max-min mapping function and takes the following form as shown in Eq. (4).

$$X_n = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (4)$$

Where X_n is the normalized data, $X_{\min}$ and $X_{\max}$ are the minimum and maximum values of each input variable, respectively, and X is the original dataset to be rescaled. The main advantage of data rescaling is that it speeds up computation while improving the precision and stability of the prediction model.

5. Statistical analysis and K fold cross-validation study

The most common method to validate the adopted ML algorithm is by using k-fold cross-validation. The k represents the number of folds that the adopted dataset is divided for the training and testing purpose (usually 10 fold is mostly used in literature) and is shown in Fig. 9. For this, the adopted dataset was separated into 10 divisions randomly, and one of the test data for the experiment was taken among the 10 data set. Within the optimum computational time, the 10 K-fold validation strategy offers trustworthy variance (Kohavi, 1995). In each of the ten rounds of model building and validation, a distinct data subset was accepted along with additional data subsets for training and testing. The algorithm's accuracy in 10 rounds of validation for 10 models was provided as an average accuracy. The k-fold cross-validation was adopted as it enhances the performance of different adopted ML models during the training and further reduces the chance of dropping the essential datasets. The cross-validation methodology adopted in the current study comprised training and testing datasets equal to 80% and

20% of total data points, respectively. It should be highlighted that for assuming ML algorithms, the k-fold cross-validation with different k was used to identify the appropriate k with higher performance.

Moreover, the output of the model was investigated based on several performance indices like mean absolute error (MAE), mean square error (MSE), root mean square error (RMSE), mean absolute percentage error (MAPE), coefficient of correlation (R), coefficient of determination (R^2), and is shown in Eq. (5), Eq. (6), Eq. (7), Eq. (8), Eq. (9), and Eq. (10) respectively. The level of linear correlation among the estimated and experimental values is indicated by R^2 .

$$MAE = \frac{1}{n} \sum_{i=1}^n |EV_i - PV_i| \quad (5)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (EV_i - PV_i)^2 \quad (6)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (PV_i - EV_i)^2}{n}} \quad (7)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{EV_i - PV_i}{EV_i} \right| \quad (8)$$

$$R = \frac{\sum_{i=1}^n (EV_i - EV_{\text{mean}})(PV_i - PV_{\text{mean}})}{\sqrt{\sum_{i=1}^n (EV_i - EV_{\text{mean}})^2 \sum_{i=1}^n (PV_i - PV_{\text{mean}})^2}} \quad (9)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (EV_i - PV_i)^2}{\sum_{i=1}^n (EV_i - PV_{\text{mean}})^2} \quad (10)$$

Where PV_i and EV_i is the predicted and measured/experimental value of the variable (CS), respectively, n is the number of data, and PV_{mean} and EV_{mean} is the average model output value representing the CS and experimental value, respectively.

A model is often regarded as accurate and meaningful when its R^2 value is more than 0.8. The more closely a model's R^2 score approaches 1, the more accurate its predictions (Gandomi et al., 2012). While MSE and RMSE simultaneously show the level of discrepancy between the tested and predicted results. The accuracy of the model's predictions improves as the MSE and RMSE values decrease. Similarly, MAE reflects the actual prediction error values (difference between tested values and predicted values). The better performance of the ML model is denoted by MAE values closer to 0. Additionally, MAPE shows the ratio of the prediction error to the tested value (Song et al., 2021), (Li et al., 2023).

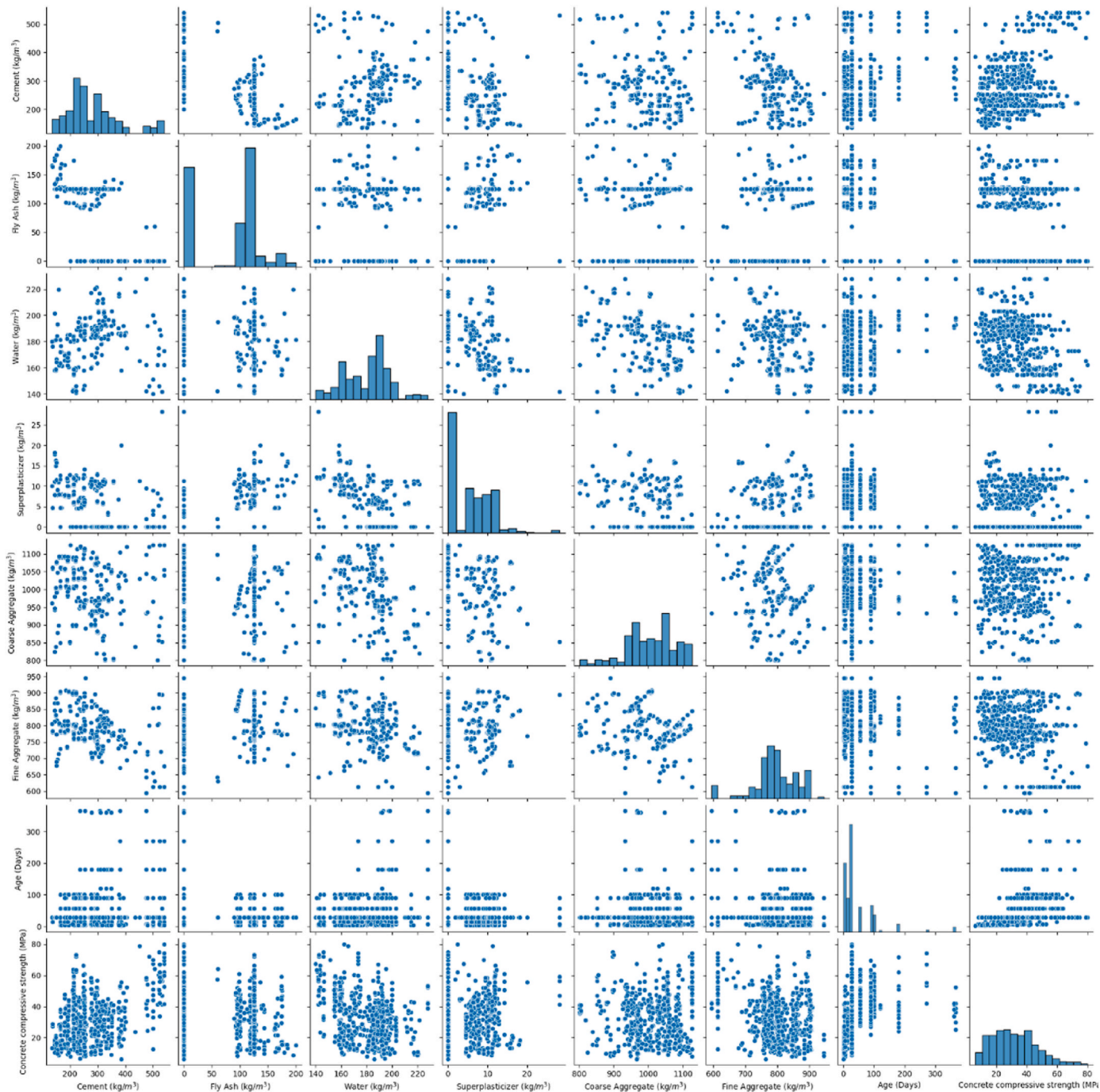


Fig. 6. Relationship of input variables with compressive strength.

Table 2
Statistical indication of parameters.

Ingredients	Count	Mean	S.D	Min	25%	50%	75%	Max
Cement (kg/m ³)	633	280.82	94.51	135	214	252	322.3	540
Fly ash (kg/m ³)	633	81.87	60.90	0	0	117.5	125.1	200
Water (kg/m ³)	633	180.56	18.31	140	165	185.7	192	228
Superplasticizer (kg/m ³)	633	5.45	5.36	0	0	5.7	9.9	28.2
Coarse aggregate (kg/m ³)	633	1004.86	72.05	801	961	1006	1056	1125
Fine aggregate (kg/m ³)	633	794.77	67.99	594	762	795	847	945
Age (Days)	633	44.50	58.83	1	14	28	56	365
Strength (MPa)	633	31.78	14.39	6.27	20.8	30.23	41.22	80

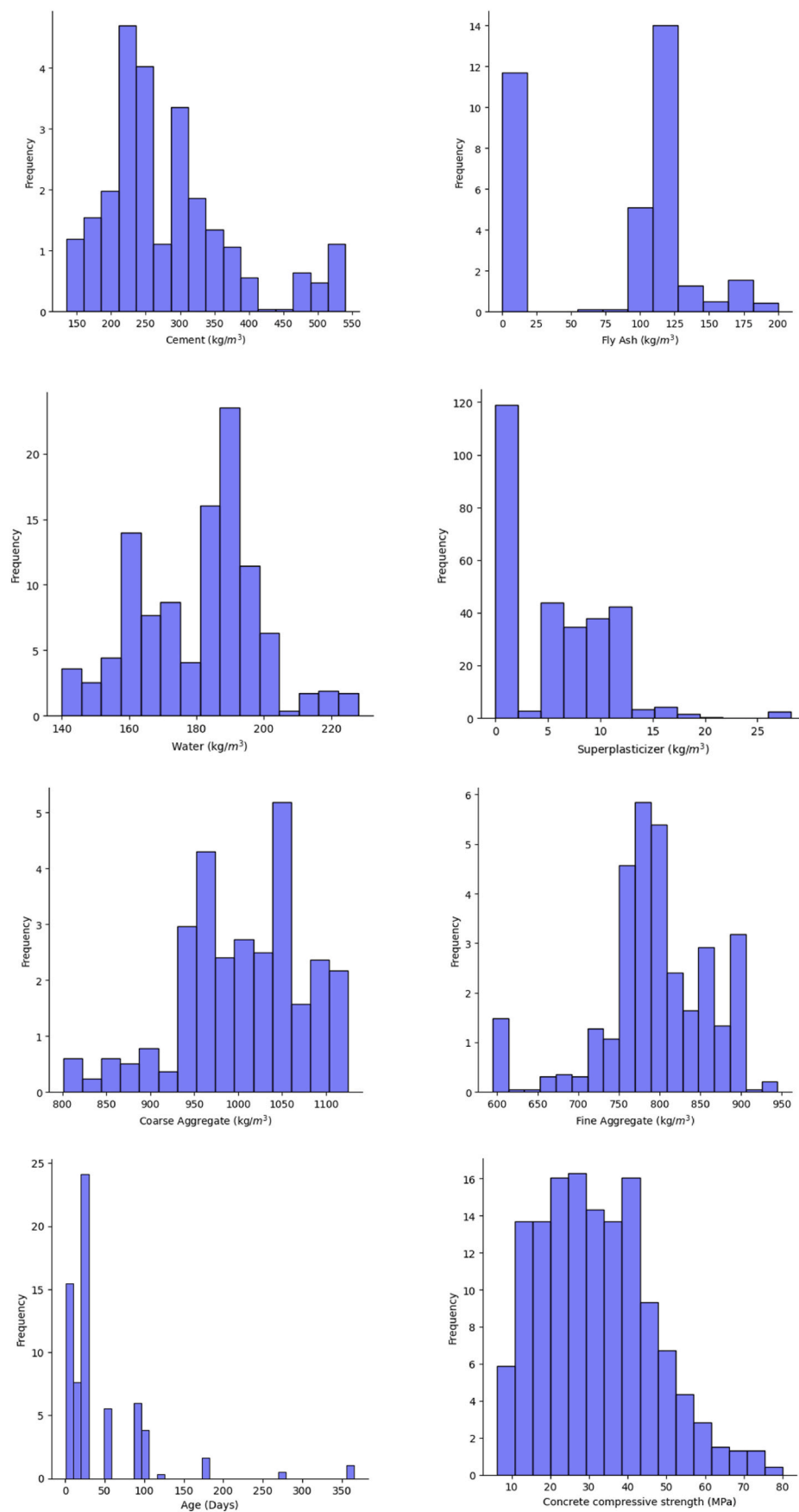


Fig. 7. Relative frequency distribution of input variables.

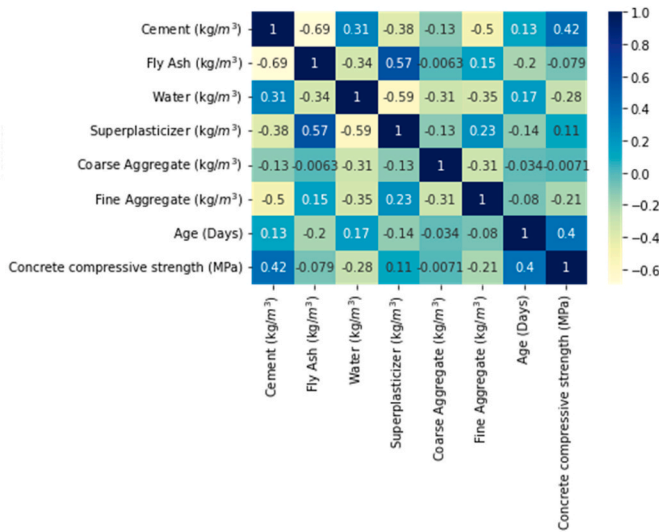


Fig. 8. Pearson's correlation coefficient between two variables.

Hyperparameters are a set of parameters that are adopted to control the behavior of the algorithm/model while obtaining a set of hyperparameter values and results in an improvised model with optimal performance. The hyperparameter combination minimizes the predefined loss function and improves the performance of the model to produce better results with fewer errors. In this study, hyperparameter tuning was adopted for the AdaBoost regressor, and the XGBoost regressor was tuned with increasing search space.

It is difficult to evaluate two forecasting models when they each have a different slope value and varied R-value. In view of this, the a20-index, a new engineering index, has recently been suggested for the reliability assessment of the created ML techniques, as shown in Eq. (11).

$$a20 - index = \frac{m20}{M} \quad (11)$$

where, M is the total sample count for the dataset, and m20 is the total sample count for samples having an experimental value/predicted value ratio between 0.80 and 1.20. Be aware that the values of the a20-index are anticipated to be the unit value for an ideal prediction model. The proposed a20-index has the benefit of having a physical engineering meaning because it indicates the percentage of samples that meet the expected values with a deviation of 20% from experimental values (Asteris et al., 2021b).

This study demonstrates how different ML approaches are reflected from a prediction perspective. It helps engineering researchers better grasp how choosing the right input parameters and regressor before running the model will provide results that are as planned. The relevance of using models in the field of civil engineering is indicated by the accuracy level of those models, particularly when predicting the

strength qualities of concrete because, in practice, determining the concrete's strength is a time-consuming task. By reducing the time and physical activities, this strategy is currently fostering the scientific community. Additionally, the performance of non-ensemble ML algorithms like Multiple LR and SVR and ensemble ML algorithms like AdaBoost regressor, Random Forest regression, XGBoost regressor, and Bagging regressor were compared as discussed in the subsequent section.

6. Results and discussion of the model

6.1. Multiple Linear regressions (MLR)

The comparison of actual (experimental) and predicted (based on ML models) values with the MLR model is shown in Fig. 10 (a). Moreover, the distribution of error of the predicted values from the experimental values is shown in Fig. 10 (b). The MLR model exhibited lower accuracy with an adjusted R² value of 0.49.

The MAE and MSE were obtained to be equal to 7.68 MPa and 88.43 MPa², respectively, showing a high error distribution. Hence, the linear regression model can be used to partially predict the outcomes of nonlinear analysis outcomes.

6.2. Support vector regression (SVR)

The comparison results from the tests and predictions with the SVR model are shown in Fig. 11 (a). Moreover, the distribution of error of the predicted data from the experimental values is shown in Fig. 11 (b).

The SVR model exhibited lower accuracy with an adjusted R² value of 0.84. It was observed that with the use of SVR, the model performance was considerably improved as compared to the MLR model. Moreover, it should be mentioned herein that the parameter tuning enhanced the performance of the ML model significantly. There are several techniques for parameter tuning, such as random search, grid search, and Bayesian optimization. The objective of these techniques is to obtain the best set of hyperparameters that results in the highest model performance. In the current study, the SVR model exhibited MAE, MSE, and R² equal to 7.20, 81.99, and 0.55 before tuning and 3.76, 27.06, and 0.85 after hyperparameter tuning with increased search CV, respectively. The increase in the value of the R² and decrease in the error distribution resulted due to the application of hyperparameter tuning.

6.3. AdaBoost

The comparison results from the tests and predictions with the AdaBoost model are shown in Fig. 12 (a). Moreover, the distribution of error of the estimated values from the experimental results is shown in Fig. 12 (b).

The MAE and MSE were obtained to be equal to 3.42 MPa and 20 MPa², respectively, showing a low error distribution. Moreover, the AdaBoost model exhibited higher accuracy with an adjusted R² value of 0.89 as compared to non-ensemble MLR and SVR machine learning

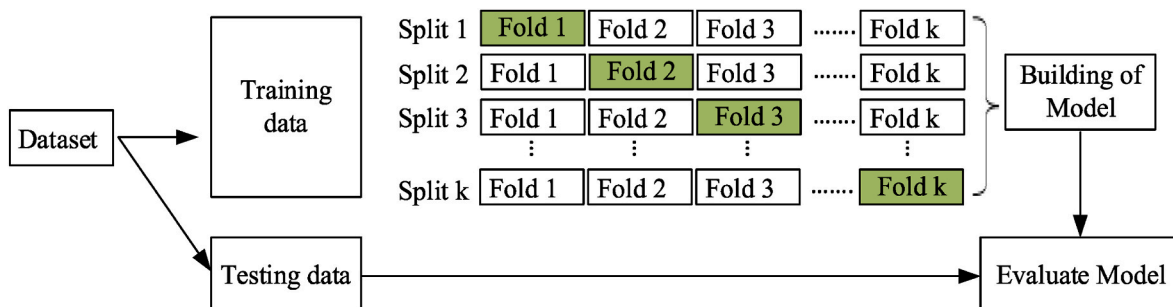


Fig. 9. Schematic representation of k-fold cross-validation.

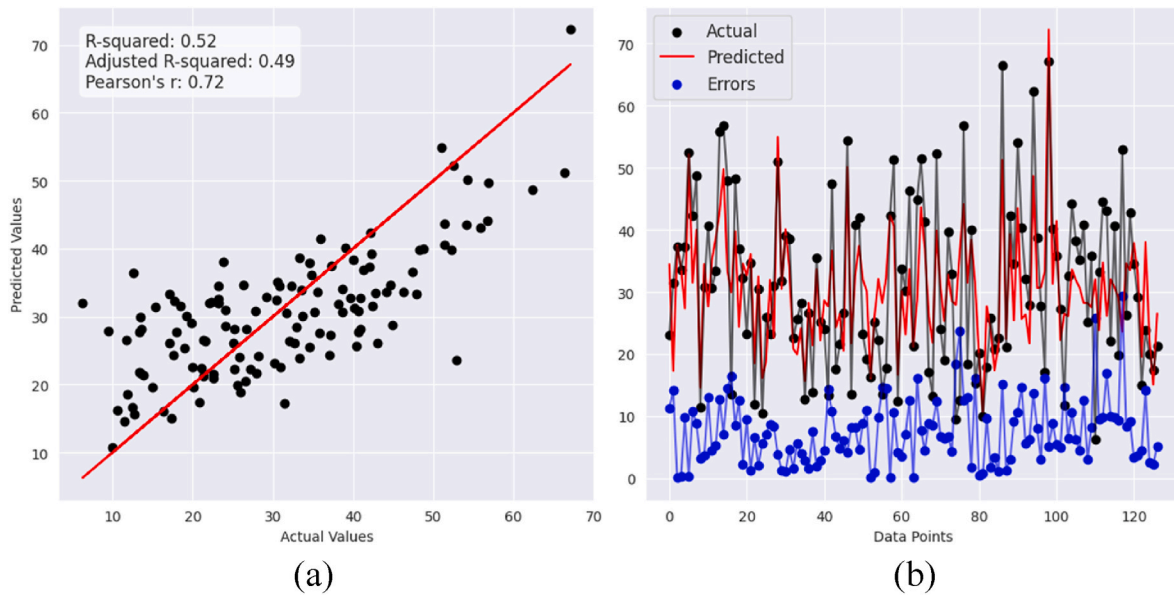


Fig. 10. (a) Prediction based on MLR; (b) Distribution of tested and predicted error.

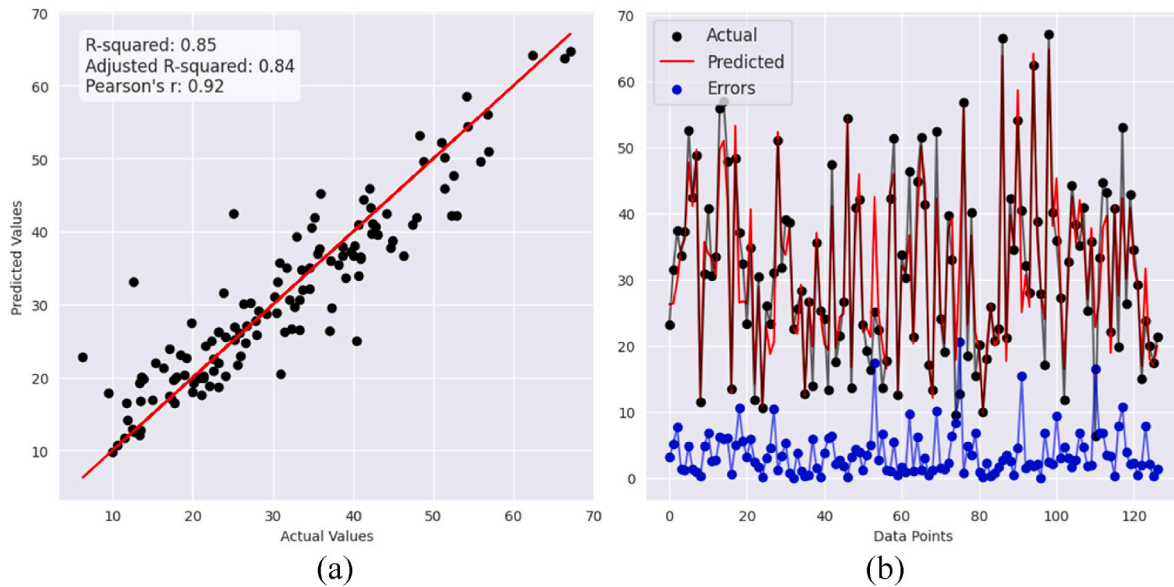


Fig. 11. (a) Prediction based on SVR; (b) Distribution of tested and predicted error.

models. The implementation of ensemble approaches can improve the performance of predicted models and improve accuracy, as evidenced by the lower levels of errors and higher values of the coefficient of determination (R^2). In general, a low R^2 score indicates that the model is not able to explain much of the variance in the data. This can be due to a variety of factors, including underfitting, overfitting, or the model being poorly suited to the data. Also, the AdaBoost model exhibited MSE and R^2 equal to 32.52 and 0.822 before tuning and 20 and 0.89 after hyperparameter tuning with grid search CV, respectively.

6.4. Bagging

The comparison of test and estimated values with the ensemble Bagging model is shown in Fig. 13(a). Moreover, the distribution of error of the predicted values from the experimental results is shown in Fig. 13(b). The Bagging model exhibited improved accuracy with an adjusted R^2 value of 0.90.

Also, the MAE and MSE were obtained to be equal to 2.91 MPa and 16.63 MPa², respectively, showing a low error distribution. Also, the Bagging model exhibited MAE, MSE, and R^2 equal to 2.99, 18.48, and 0.89 before tuning and 2.91, 16.63, and 0.909 after hyperparameter tuning with grid search CV, respectively.

6.5. Random forest (RF) regression

The assessment of actual and estimated values with the ensemble RF regression model is shown in Fig. 14(a). Moreover, the distribution of error of the predicted values from the experimental values is shown in Fig. 14(b).

The RF regression model's adjusted R^2 score of 0.92 demonstrated greater accuracy. Additionally, it was determined that the MAE and MSE were equivalent to 2.86 MPa and 15.99 MPa², respectively, demonstrating the low error distribution.

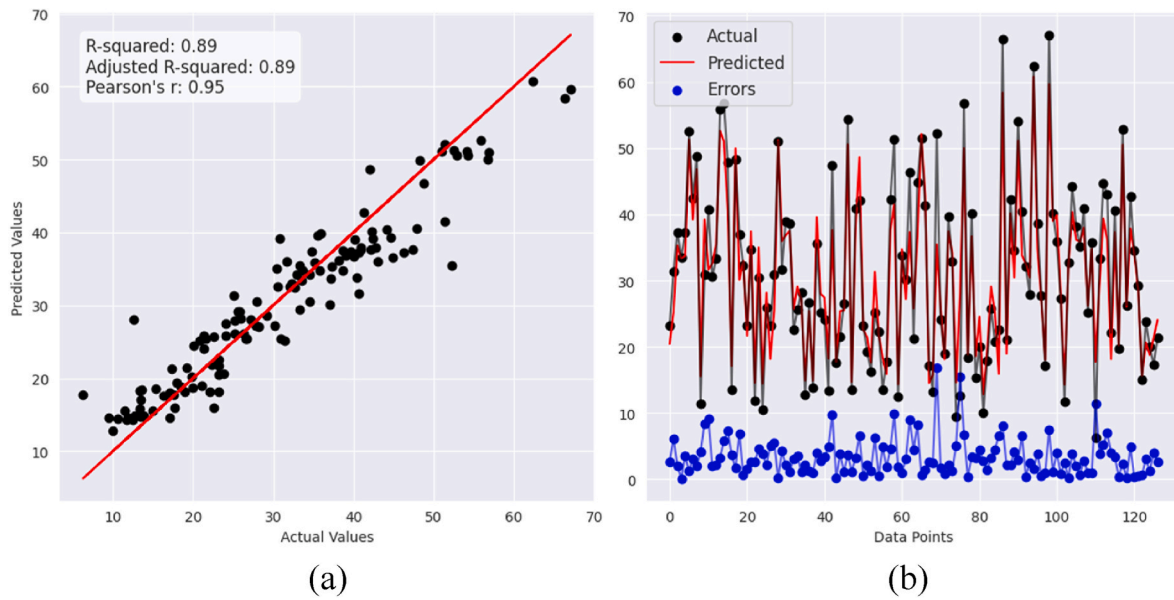


Fig. 12. (a) Prediction based on AdaBoost; (b) Distribution of tested and predicted error.

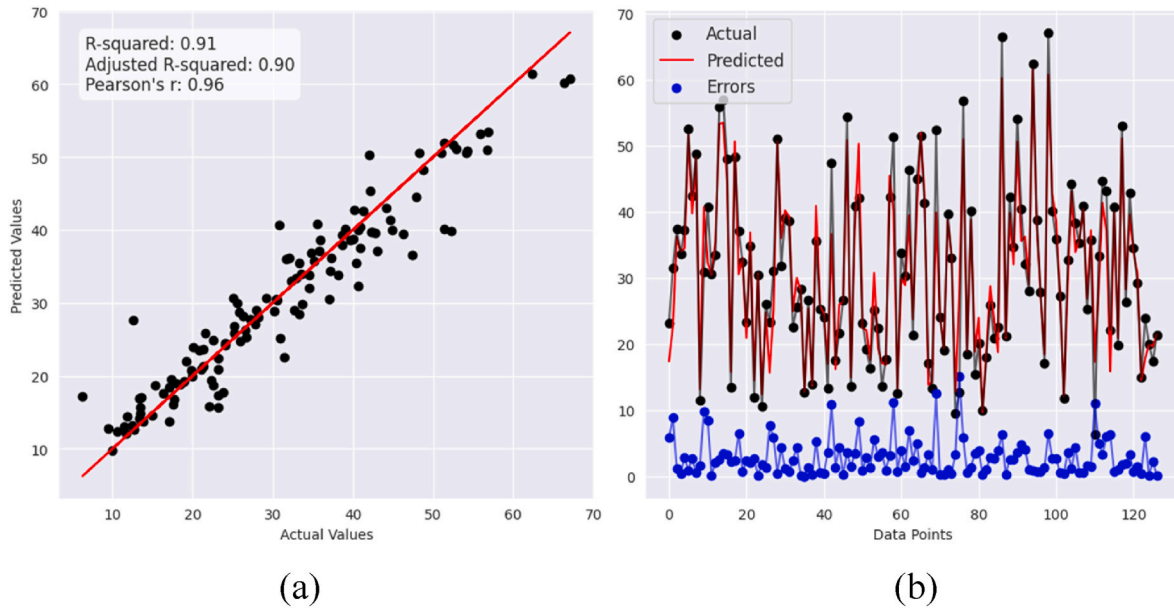


Fig. 13. (a) Prediction based on Bagging; (b) Distribution of tested and predicted error.

6.6. XGBoost

The comparison of actual (experimental) and predicted (based on ML models) values with the XGBoost model is shown in Fig. 15 (a). Moreover, the distribution of error of the predicted values from the experimental values is shown in Fig. 15 (b). The XGBoost model exhibited the highest accuracy with an adjusted R^2 value of 0.95.

XGBoost regressor model exhibited MAE, MSE, and R^2 equal to 2.55, 14.29, and 0.922 before tuning and 2.13, 9.36, and 0.95 after hyperparameter tuning with increasing random search CV, respectively. Parameter tuning is important for XGBoost for performance optimization, overfitting prevention, model generalization, etc. The performance of the XGBoost model can be considerably enhanced with the appropriate choice of hyperparameters. The rate of convergence and the accuracy of the model's predictions can be influenced by hyperparameters like learning rate, maximum depth, and subsample rate. If the

complexity of the model is too great, XGBoost is prone to overfitting. Controlling the model's complexity and avoiding overfitting can both be accomplished using parameter adjustment. Finding the optimal set of hyperparameters that enables the model to generalize successfully on fresh, untested data is made possible via parameter tuning.

These comparison charts show that the ensemble learning models are significantly better at making predictions than the individual non-ensemble regression models. Furthermore, the statistical analysis also reveals the robustness of the models that were used. The estimation of evaluation metrics for the studied ML algorithms is shown in Table 3. Lower RMSE and MAPE values and higher R values are characteristics of effective models with strong prediction abilities. Additionally, for the perfect predictive model, the value of a20-index should be equal to 1. The value of the a20-index of XGBoost was the best among the studied ML algorithms, as shown in Table 3.

Table 3 shows that the XGBoost regressor is the most accurate model

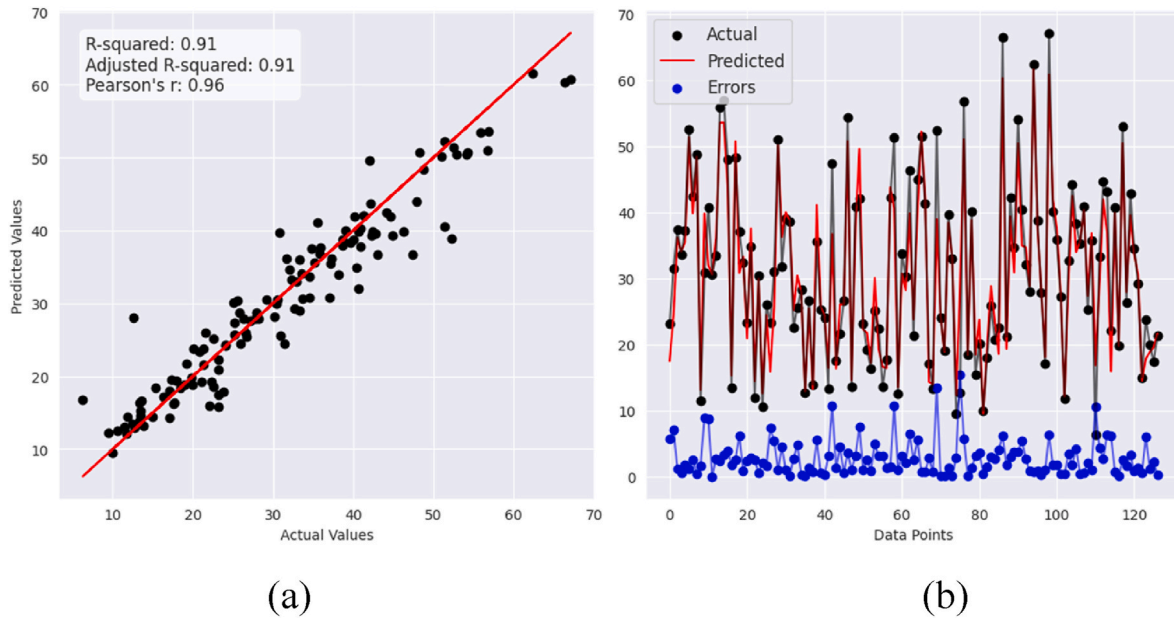


Fig. 14. (a) Prediction based on RF regression; (b) Distribution of tested and predicted error.

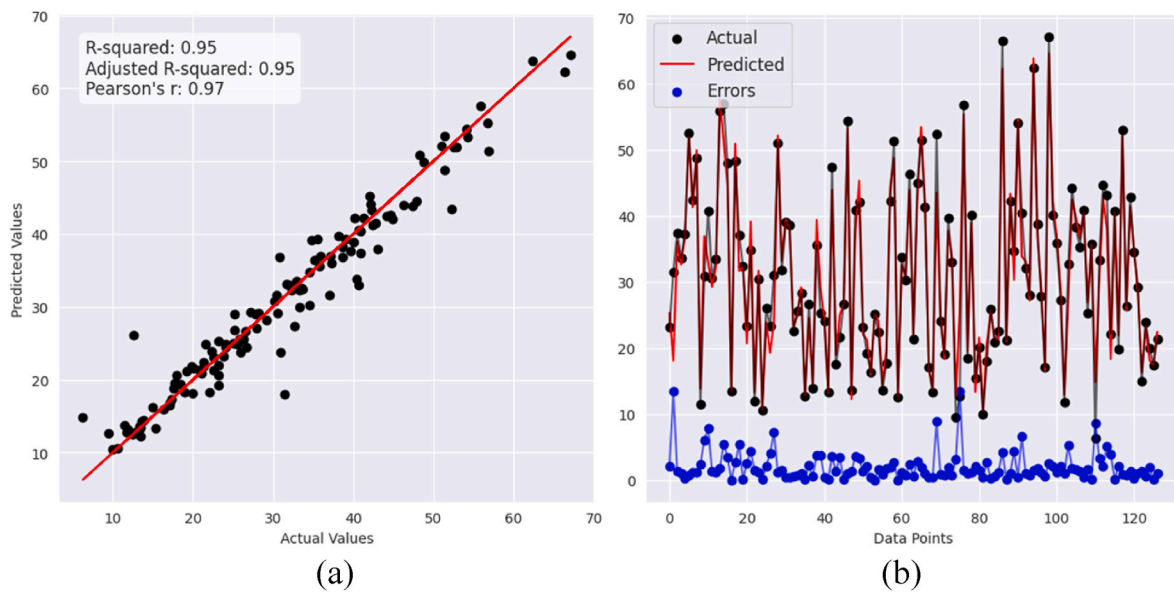


Fig. 15. (a) Prediction based on XGBOOST; (b) Distribution of tested and predicted error.

Table 3

Estimation of evaluation metrics for the studied ML algorithms.

Algorithm	MAE (MPa)	MSE (MPa ²)	R ²	MAPE	RMSE (MPa)	a20- index
Multiple LR	7.68	88.43	0.520	33.17	9.40	0.433
SVR (Tuning)	3.76	27.06	0.850	16.01	5.20	0.818
AdaBoost regressor	3.42	20.0	0.890	13.90	4.47	0.819
Bagging regressor	2.91	16.633	0.910	11.58	4.078	0.850
Random forest regressor	2.86	15.99	0.910	11.29	3.99	0.858
XGBoost (tuning)	2.13	9.36	0.950	8.177	3.06	0.913

having the highest value of R^2 equal to 0.96, while the least accuracy was obtained by the Multiple LR model with R^2 equal to 0.52. All the models other than multiple LR can be considered highly reliable predictive models, i.e., models with R^2 values more than 0.85 (Kioumars et al., 2023).

6.7. K fold cross-validation and statistical analysis

Prediction models must achieve the appropriate level of accuracy to be valid. By randomly rearranging the given data, the 10 K-fold cross-validation approach was performed to make sure the model is accurate. This method reduces the bias brought on by a random sampling of the training data set. The test data sets were divided into 10 equal subsets using this technique, and the nine subsets were used to construct and shape the strong learner. The final subset, meantime, was used to evaluate the applicability of the created model. The process of validation

was repeated in 10 numbers, and the average accuracy was determined at the end of the repetition. By adopting k-fold cross-validation, the risk of overfitting the model to a single train-test split was decreased while obtaining a more precise estimate of the model's performance. Cross-validation is mostly used to optimize the model. In the current study, the scores seem to be better, with the number of splits equal to 5. The regression results (R^2) and Statistical error (MSE, MAE, and RMSE) result after 10 k-fold cross-validations are shown in Fig. 16.

The findings demonstrate that the model's performance is improved by the adoption of ensemble methods, going from a weak to a strong relationship and yielding decisive conclusions. The accuracy and dependability of the relevant created models are demonstrated by the results of the 10 K-fold cross-validation procedure. For the 10 K-fold validation of various ML approaches, there is fluctuation in the value of R^2 , but each fold still maintains a high level of accuracy. For example, the range of R^2 for MLR, SVR, AdaBoost regressor, Bagging regressor, Random forest regressor, and XGBoost is 0.52–0.59, 0.82–0.86, 0.84–0.89, 0.84–0.91, 0.84–0.91, and 0.89–0.95, respectively. Also, the average value of statistical error tool MSE for MLR, SVR, AdaBoost regressor, Bagging regressor, Random forest regressor, XGBoost is 82.73 MPa, 30.95 MPa, 23.74 MPa, 21.22 MPa, 20.83 MPa, and 13.15 MPa, respectively. The purpose of using k-fold cross-validation is to obtain a more precise approximation of the performance of the model and to decrease the risk of overfitting the model to a single train-test split. Hence, the results from the k-fold cross-validation method depict the reliability and accuracy of the studied algorithms.

6.8. Model comparison

The various compositions of cement, fine aggregate (FA), coarse aggregates (CA), fly ash (FA), water content, percentage of super-plasticizer, and curing days, results in different the CS of concrete. In other words, it is commonly known that different materials have different compositions depending on their compressive strength range (Ali et al., 2022). Hence, further research work was carried out to evaluate the model performance in different strength ranges as shown in Table 4.

From Tables 4 and it was observed that the XGBoost model performed the best, especially when it came to predicting the CS of the concrete with FA and had a strength between 31 and 57 with R^2 equal to 0.737. However, the application of the model is more accurate in the 6–31 MPa range due to the bigger size of the data points; the R^2 score for this range was 0.721.

6.9. Sensitivity and parametric analysis

The sensitivity and parametric analysis can help to further analyze the data type and explore the importance/contribution of each input feature for the corresponding output (Feng et al., 2020), (Al-Hashem et al., 2022), (Kioumars et al., 2023), (Amin et al., 2021), (Shen et al., 2022). The feature importance/sensitivity analysis was performed following the model optimization. The study helps to understand and identify the influence of each feature/component of concrete during the concrete compressive strength predictions. The feature importance of

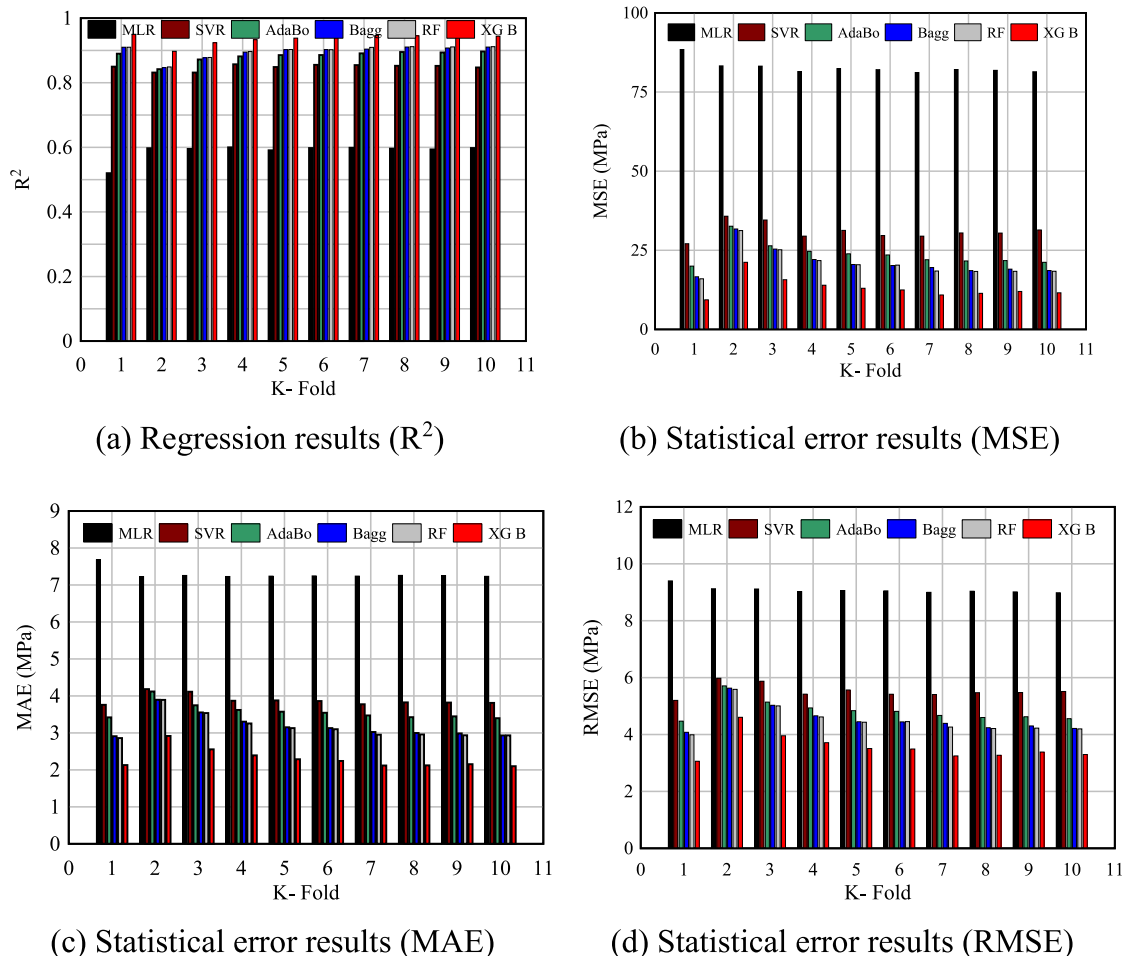


Fig. 16. (a) Regression results (R^2) and Statistical error (MSE, MAE, and RMSE) results after 10 k-fold cross-validations.

Table 4

Comparison of studied ML algorithms for different ranges of strengths.

Algorithm	CS range (MPa)	No. of data	MAE (MPa)	MSE (MPa ²)	R ²	MAPE	RMSE (MPa)	a20-index	Remarks ^a
Multiple LR	6–31	324	4.349	25.775	0.396	26.719	5.076	0.569	Poor
	31–57	275	4.099	26.835	0.369	10.129	5.180	0.891	Poor
	57–80	34	5.332	37.674	0.537	7.95	6.137	1	Poor
SVR (Tuning)	6–31	324	2.628	14.118	0.669	14.806	3.757	0.785	Good
	31–57	275	3.114	21.354	0.498	7.547	4.621	0.909	Poor
	57–80	34	4.845	36.56	0.492	6.991	6.047	1	Poor
AdaBoost regressor	6–31	324	2.365	11.546	0.729	13.761	3.398	0.785	Good
	31–57	275	2.865	15.060	0.646	7.044	3.881	0.964	Good
	57–80	34	3.999	19.759	0.194	5.943	4.445	1	Poor
Bagging regressor	6–31	324	2.44	12.379	0.709	14.425	3.518	0.785	Good
	31–57	275	2.912	14.541	0.659	7.094	3.813	0.98	Good
	57–80	34	4.559	32.95	0.344	6.637	5.74	1	Poor
Random forest regressor	6–31	324	2.418	12.027	0.718	14.649	3.467	0.785	Good
	31–57	275	3.116	17.156	0.549	7.473	4.142	0.96	Good
	57–80	34	4.289	22.111	0.0978	6.322	4.702	1	Poor
XGBoost (tuning)	6–31	324	2.418	11.88	0.7214	14.42	3.447	0.831	Good
	31–57	275	2.114	9.986	0.737	5.194	3.16	0.98	Good
	57–80	34	4.567	31.921	0.302	6.596	5.649	1	Poor

^a The model performance is “good” when R² is ≥ 0.5 and it is “poor” when R² is < 0.5 .

the concrete strength prediction model is shown in Fig. 17.

It was observed that age is the utmost essential feature, followed by cement and water with respective information gain values of 32.91, 23.50 and 15.10. The influence of cement content and age is dominant in predicting the CS of concrete, which is in line with the practice in engineering. The aggregate, fly ash, and superplasticizer were observed to have a lower effect on strength prediction being the least important feature. Age, cement, water, fine aggregate, fly ash superplasticizer, and coarse aggregate have their contribution to the compressive strength of concrete but in decreasing order. The compressive strength of concrete is proportional to the seven variables, as can be seen from the importance score of the variables, meaning that a change in any one of the seven variables will alter the CS of concrete. However, engineers should focus more on age and cement when designing concrete and less on the amount of aggregate, fly ash, and superplasticizer because age and cement have a significant impact on the CS of concrete (Li et al., 2023), (Feng et al., 2020).

The sensitivity analysis was further carried out as a parametric analysis to authenticate the reliability of the adopted AI model. Then, the performance was evaluated based on the simulated dataset. This standardized approach was also employed for the remaining input variables in the study. Table 5 illustrates the dataset utilized for the additional parametric analysis.

The effect of most essential features was studied based on the parametric study to determine their contribution to predicting the CS, as shown in Fig. 18. The XGBoost was adopted for the parametric analysis

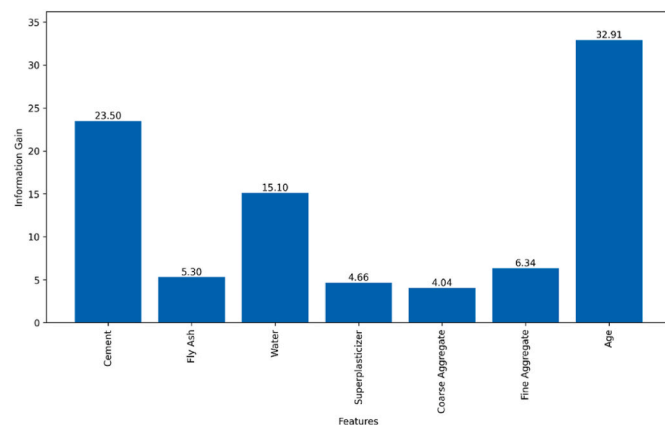


Fig. 17. Influence of input parameters (information gain) for the estimation of output.

Table 5

Parametric analysis data.

Input parameters		Other Variables
Parameter	Range	
Age	1–365	C = 280.81, Sp = 5.44, Fly = 81.86, C. agg. = 1004.85, F. agg. = 794.77, Water = 180.56
Cement	134–540	Age = 44.5, Sp = 5.44, Fly = 81.86, C. agg. = 1004.85, F. agg. = 794.77, Water = 180.56
Fly ash	0–200	Age = 44.5, C = 280.81, Sp = 5.44, C. agg. = 1004.85, F. agg. = 794.77, Water = 180.56

due to its superior performance as compared to all of the adopted non-ensemble and ensemble algorithms. The study was conducted by systematically controlling the input parameter within its minimum and maximum values available in the dataset while simultaneously maintaining a consistent average for the other parameters. With an increase in cement content, the strength was observed to increase, which is practically feasible.

The increase in age and fly ash increased the strength of concrete until some range, and later on, the strength was observed to remain constant. These observations follow the studies conducted by past researchers (Al-Hashem et al., 2022), (Wilińska and Pacewska, 2018). From Fig. 18., it is evident that the strength of concrete increased rapidly within 28 days, and it is slower up to the first 100 days, then the strength does not vary at later ages. A similar result was observed in the case of fly ash since concrete strength increased for 80 kg/m³ fly ash content, beyond which its effect on strength is negligible.

7. Limitations of this research

Even though this study is based on a scientific investigation of many input factors to predict the compressive strength of concrete, it contains several shortcomings that need to be addressed. These key restrictions are described below: Scientific techniques like Non-ensemble: Multiple LR and SVR and Ensemble: AdaBoost regressor, Random forest regression, XGBoost regressor, and Bagging regressor involve training and testing data based on the models' performance. A big precise concrete change with a fly ash index may be utilized to further explore the performance of the model with several other input parameters. Our work is limited by the fact that the dataset used to develop our prediction models was very small; a larger dataset would have improved the models' accuracy and dependability.

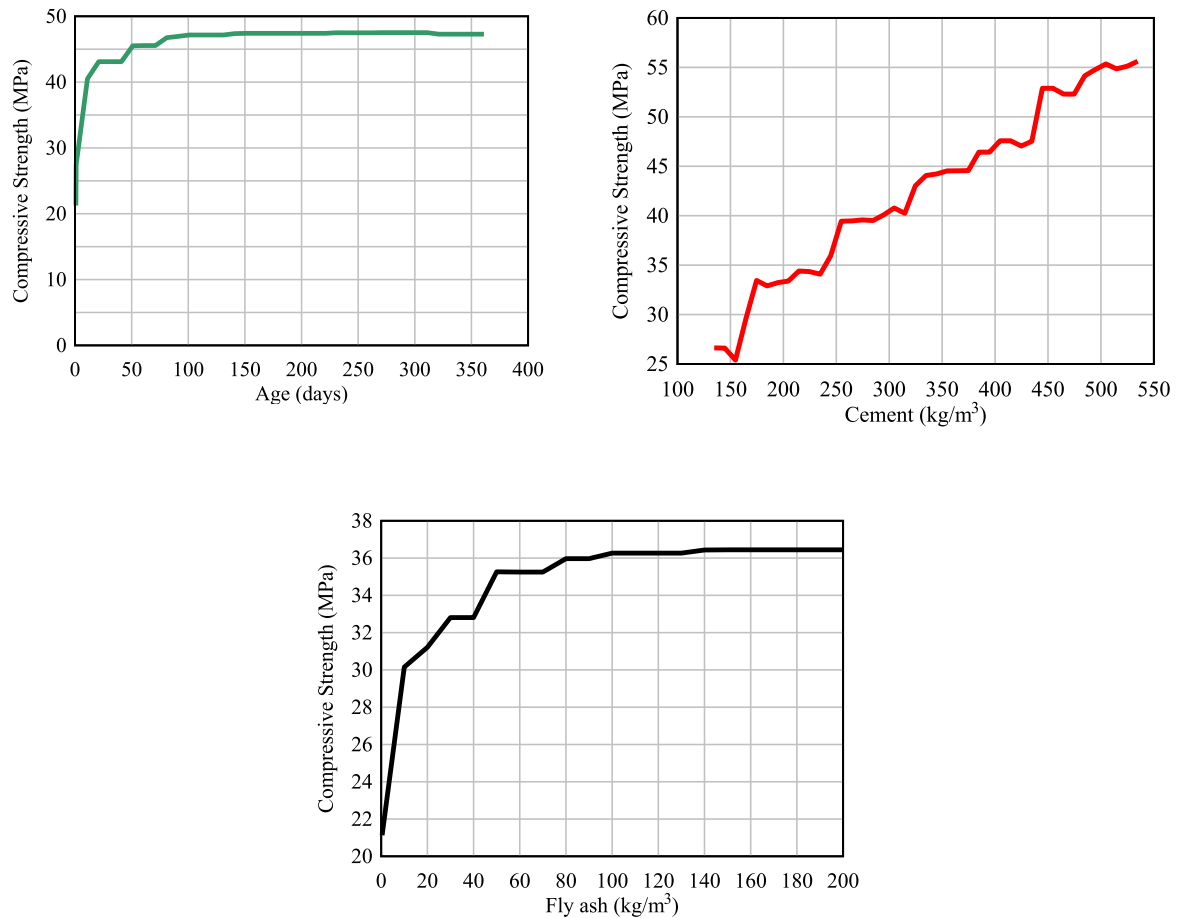


Fig. 18. Parametric analysis to explore the effect of age, cement, and fly ash on the prediction of CS of concrete.

8. Conclusion

In this research, a comprehensive analysis of 633 data sets on the compressive strength (CS) of fly ash-based concrete was conducted using information gathered from past journal papers (Song et al., 2021), (Song et al., 2022), (Mahajan and Bhagat, 2022). The primary objective of this study was to compare the performance of non-ensemble and ensemble-selected machine learning (ML) models in predicting the CS of concrete containing fly ash. The non-ensemble ML algorithm like Multiple LR and SVR and ensemble ML algorithms like AdaBoost regressor, Random Forest regression, XGBoost regressor, and bagging regressor were employed. The results obtained from both types of models were compared to assess their predictive capabilities. The following key findings emerged from this study.

1. The substantial evidence in favor of using AI approaches to predict the compressive strength of concrete containing fly ash comes from the good agreement between the predicted and experimental results. All the models other than multiple LR can be considered highly reliable predictive models, i.e., models with R^2 values more than 0.85.
2. The ensemble ML models demonstrated superior performance, as evidenced by higher values of R^2 and lower errors (MSE and MAE), or reduced variance between the actual and predicted outcomes.
3. The comparison reveals that XGBoost Regressor is the most reliable model, demonstrating the highest coefficient of determination (R^2) of 0.95, a-20 index of 0.913 and the lowest RMSE value of 3.06 MPa and MAE of 2.13 MPa, while the Multiple LR model is the least

accurate method with R^2 equal to 0.516, a-20 index of 0.433, RMSE of 9.40 MPa and MAE of 7.68 MPa.

4. On the basis of sensitivity analysis, age of the concrete is the most influential feature in predicting compressive strength, followed by cement and water, with information gain values of 32.91, 23.50, and 15.10, respectively. On the other hand, aggregate and fly ash were observed to have a comparatively lower impact on strength prediction, indicating their lesser importance. Consequently, engineers should prioritize the consideration of age and cement content when designing concrete structures while allocating less emphasis on the amount of aggregate, fly ash, and superplasticizer, as age and cement exhibit significant influences on the compressive strength of concrete.
5. XGBoost regressor model exhibited MAE, MSE, and R^2 equal to 2.55, 14.29, and 0.922 before tuning and 2.13, 9.36, and 0.95 after hyperparameter tuning, respectively. Similarly, the SVR model exhibited MAE, MSE, and R^2 equal to 7.20, 81.99, and 0.55 before tuning and 3.76, 27.06, and 0.85 after hyperparameter tuning. Hence, parameter tuning is important for performance optimization, addressing overfitting and model generalization.

9. Suggestions and recommendations

1. To forecast the compressive strength of the concrete with Fly ash, several soft computing models should be applied.
2. It is also important to develop the impact of fly ash on the tensile and flexural strengths.
3. Experiments should be carried out to confirm the developed models.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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